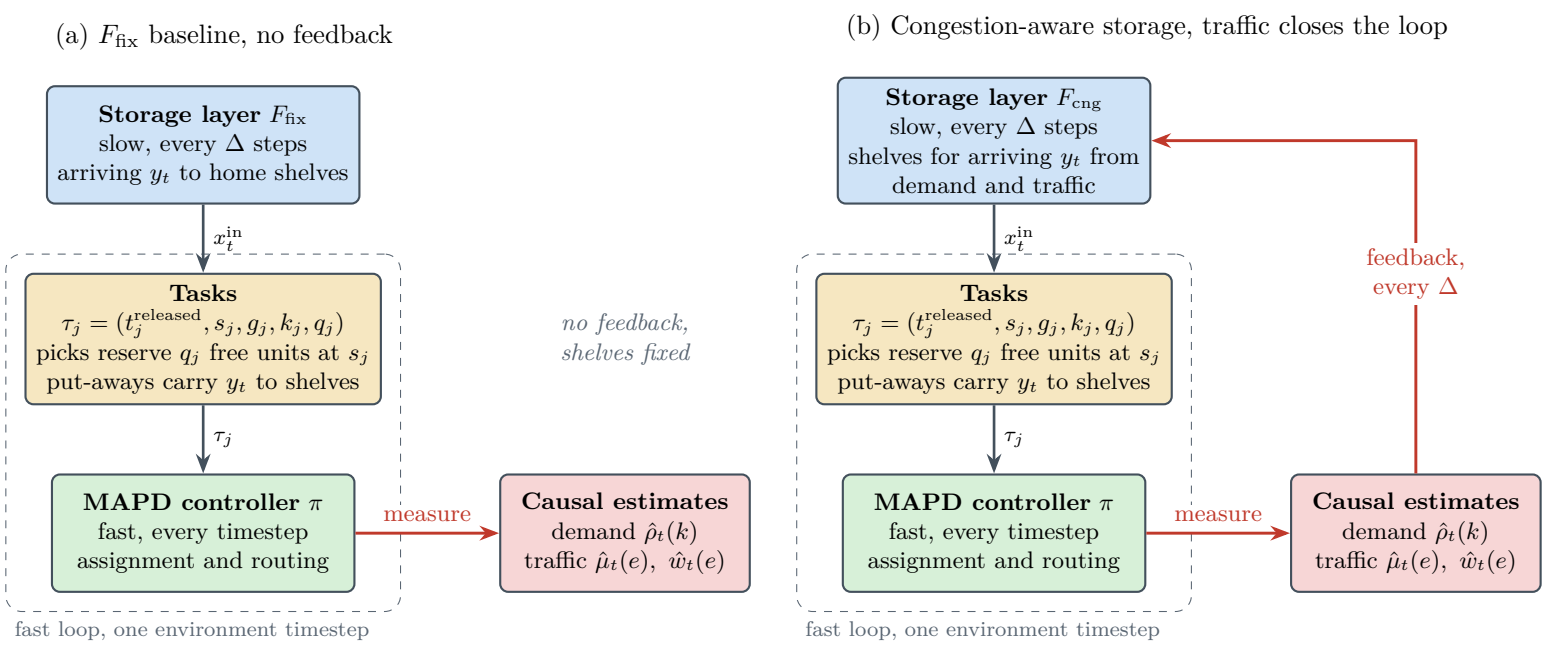

(a) F_{fix} baseline, no feedback
Storage layer F_{fix}
slow, every Δ steps
arriving y_t to home shelves
x_t^{in}
Tasks
$\tau_j = (t_j^{\text{released}}, s_j, g_j, k_j, q_j)$
picks reserve q_j free units at s_j
put-aways carry y_t to shelves
τ_j
MAPD controller π
fast, every timestep
assignment and routing
measure
Causal estimates
demand $\hat{\rho}_t(k)$
traffic $\hat{\mu}_t(e)$, $\hat{w}_t(e)$
fast loop, one environment timestep
no feedback,
shelves fixed
(b) Congestion-aware storage, traffic closes the loop
Storage layer F_{cng}
slow, every Δ steps
shelves for arriving y_t from
demand and traffic
x_t^{in}
Tasks
$\tau_j = (t_j^{\text{released}}, s_j, g_j, k_j, q_j)$
picks reserve q_j free units at s_j
put-aways carry y_t to shelves
τ_j
MAPD controller π
fast, every timestep
assignment and routing
measure
Causal estimates
demand $\hat{\rho}_t(k)$
traffic $\hat{\mu}_t(e)$, $\hat{w}_t(e)$
feedback,
every Δ
fast loop, one environment timestep